Jack Burke

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Professional Summary [Tailor to Job Posting – General Example to Build on Below]

Engineering Consultant with 5 years of excellence in quickly learning about and solving complex problems through knowledge sharing, independent research, and thorough data analysis. Proven track record of managing multiple projects under tight timelines to deliver client-ready presentations, documents, and novel analytical solutions. Seeking to leverage deep analytical and project management expertise to deliver creative solutions.

Technical Skills & Experience [Move around depending on Job Application]

Data Analysis: Excel (including VBA), Web scraping (Octoparse), Coding (Java and Python)
Management: Workflow management, junior staff mentoring, conducting interviews, client relationships
Engineering Modeling and Analysis: Third Party Testing, Refrigeration Cycle Analysis (Refprop), Heat Exchanger Design (Coil Designer), CFD Simulations (Flow 3D), Control System Design (LabView), Electronic Design (EAGLE)
Building Energy Modeling: Heating/Cooling Load Calculations (Manual J), BPI Compliant Building Modeling Software (SnuggPro), ASHRAE 140 Compliant Building Modeling Software (eQuest and QBAT), ASHRAE Level II Audits
CAD Software: Creo, SolidWorks, Fusion 360

Work Experience

Guidehouse: Consultant (*2021 – 2023*); Senior Consultant: (*2023 – Current*)

Public Document Contributions:
Commercial Refrigeration Test Procedure and Energy Conservation Standard Rulemakings
Ice Maker Test Procedure and Energy Conservation Standard Rulemakings
Vending Machine Test Procedure and Energy Conservation Standard Rulemakings
Georgia Multifamily Rebate Documents

- Current Project: Distributing rebates to multifamily buildings for energy efficiency and appliance upgrades in the state of Georgia through a grant management SaaS platform (Neighborly Software).
  - Quickly (2 weeks) became the teamwide expert for multifamily building energy modeling software to provide assistance to contractors and conduct QA/QC reviews for program compliance and engineering compliance.
  - Conduct at least 20 pre- and post-install engineering reviews weekly for multifamily building upgrades, resulting in over $3 million in rebates paid to date and contributing to electrified and efficient buildings.
  - Constructed an application rebate forecast to provide monthly and total estimated rebates for the client
  - Crafted appliance installation guidelines and detailed application guidance for contractors.
  - Developed and documented internal best practices to QA/QC engineering reviews and ensure consistency and provide daily support to junior staff in reviewing applications.
  - Created a building energy modeling checklist and an energy modeling calibration tool to provide contractor support for contractors new to home energy modeling.
  - Provide daily technical support to applicants with engineering questions through our multifamily email inbox.

- Prior Project (2021-2026): Developed federal regulations and conducted engineering/data analysis for the Department of Energy's Appliance Standards Program.
  - Contributed to 6 concurrent federal rulemakings for energy conservation standards and test procedures for Commercial Ice Makers, Beverage Vending Machines, and Commercial Refrigeration Equipment; resulting in the publication of final rules in the code of federal regulations and several preliminary rulemaking publications.
  - Led bi-weekly meetings with the client (2023-2026) to provide project updates, receive feedback on deliverables, and build client trust through excellent communication of technical subjects.

- Orchestrated project workflows to revise and update the refrigeration engineering analysis model for multiple projects to account for changes in refrigeration technology; update cost/efficiency assumptions; refine excel calculations and calibration methods; and develop creative solutions to incorporate regulations from other agencies (EPA).
- Facilitated 4+ hour manufacturer interviews to collect detailed proprietary information on current energy efficiency technologies used and manufacturing facility capabilities and potential upgrade costs, which is key information to inform the engineering and manufacturer impact analysis.
- Mentored, managed, and reviewed work of junior staff to assist them in developing key skills/knowledge.
- Participated in quarterly national standard committee meetings (ASHRAE and AHRI) to provide technical and regulatory expertise to the committee resulting in the updates of several industry standards.
- Wrote test plans for third party test laboratories and reviewed test reports conducted to ensure compliance with test plans, industry test procedures, and current DOE energy standards.
- Summarized public comment feedback from efficiency advocates, manufacturers, industry trade associations, politicians, and individuals to inform later rulemaking stages with deep industry knowledge.

- Pro-Bono Project (2025): Supported the development of a strategic roadmap for a non-profit foundation
  - Compiled member feedback to inform high-priority goals in the near term (1 year) and long term (5 year) and built a dashboard to quickly analyze feedback categories.
- Prior Project (2021): Engineering analysis for a global beverage company
  - Analyzed emerging technology options and conducted cost-efficiency analysis to determine the financial viability of each design option to produce a roadmap for their beverage cooler and vending machine lineup

Liebherr-Hausgeräte Ochsenhausen GmbH: Mechanical Engineering Intern ***January 2020 – March 2020***

- Tested prototypes to increase freezing rate on ice maker while maintaining functionality requirements
- Wrote and edited standardized testing procedures to be used in in assembly plants in 5+ countries

Cooling Technologies Research Center: Undergraduate Researcher ***May 2019 – December 2019***

- Researched flexible fabrics for wearable technology applications by performing mechanical and thermal testing of materials to characterize material properties
- Published as the third author of a research paper on flexible fabrics for wearable technology in Nature.
- Wrote a technical paper and presented slides at the SURF conference for summer research.
- Conducted thermal, electrical, and optical testing to prove feasibility of a thermal switch mechanism for mobile technology applications.

Education

Purdue University – B.S. Mechanical Engineering **Graduated - May 2021; GPA: 3.86/4.00**

Minor*:* Global Engineering Studies Minor (GEARE)

Certificate: Entrepreneurship and Innovation

Capstone Project: Increasing Wave Powered Energy Production

- Researched ocean powered energy production technology to develop a feasible solution to improve energy production capacity and efficiency of "Overtopping" devices
- Self-taught Flow 3D to create CFD simulations for our CAD modeled "constructive interference overtopping device" to provide results validating our design as >10% more efficient than traditional overtopping devices